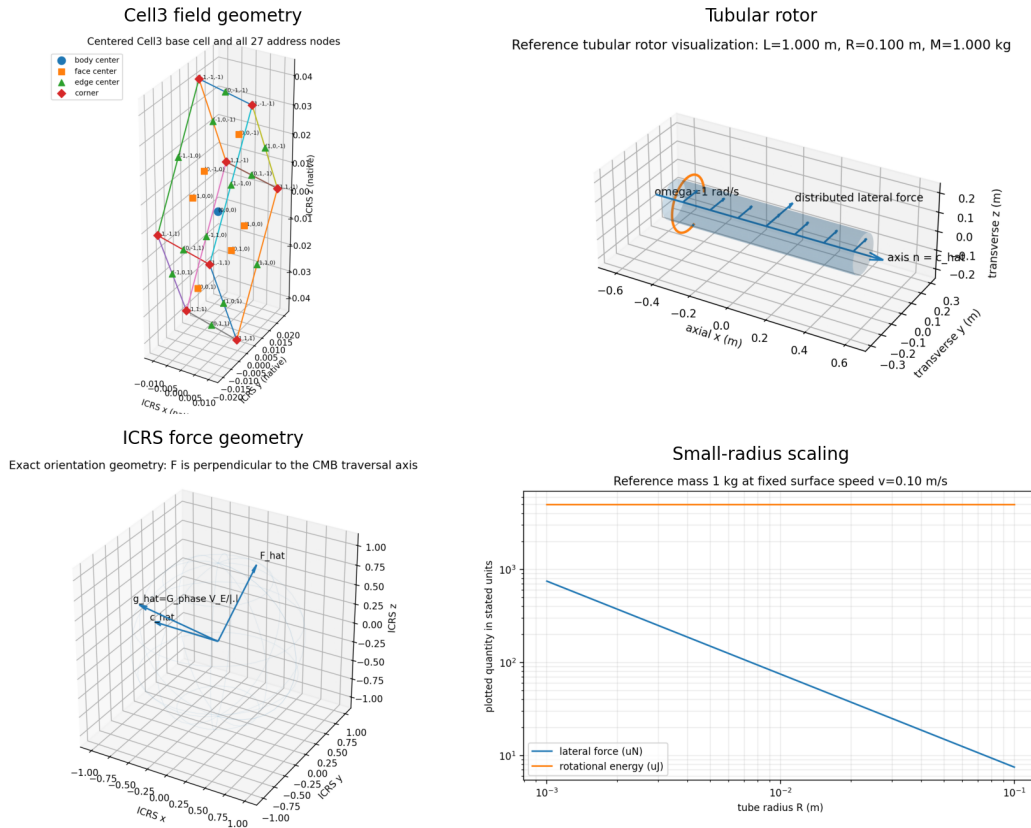


K-FIELD

Tubular Rotor Force Analysis

Continuous circumferential phase occupancy, lateral-force integration, and the long narrow rotor limit



Title figure. All panels are generated from the authoritative Cell3 geometry, calibrated Earth-CMB field inputs, exact continuous-rim integration, and the stated 1 kg reference rotor.

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This paper is a standalone educational and archival reference. It defines the Cell3 field geometry, the Earth-CMB coordinate lock, the phase-response matrix, the continuous-ring force integral, its exact extension to a uniform tubular rotor, calibrated force results, radius and speed scaling, mechanical efficiency distinctions, structural loading, and the experimentally decisive small-radius limit. Every numerical table and dimensional figure is generated from the authoritative numerical inputs or from exact computations on those inputs. Structural schematics are explicitly normalized and contain no unstated measurement values.

Abstract

A uniform rotating ring converts the phase-dependent point-mass force into a steady lateral vector because antipodal mass elements reverse both radial and tangential directions. Their two sign reversals cancel, so opposite elements add. Integrating the complete circumference gives $F = (\kappa_F M \omega / 2) \mathbf{n} \times (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$. Extending the ring axially produces a tube with exactly the same force per unit mass at the same angular speed; radius and length disappear from the field integral.

For the calibrated Earth-CMB alignment, $M=1$ kg and $\omega=1$ rad/s produce 7.505365563 uN in the fixed ICRS direction $RA=84.082114$ deg, $Dec=41.289922$ deg. The force is perpendicular to the CMB traversal axis and has no net torque for a centered, axially symmetric rotor. At fixed angular speed, shrinking radius does not change force but reduces rotational energy quadratically. At fixed surface speed, shrinking radius raises angular speed and therefore raises force per mass as $1/R$ while leaving thin-shell rotational energy unchanged.

The complete efficiency result is more precise than the statement that a long narrow tube is intrinsically best. The K-Field coupling selects small radius and complete circumferential occupancy. Cross-sectional mechanics then determines whether a hollow tube, solid rod, annulus, or segmented stack is best. A solid rod minimizes radius of gyration and stored spin energy; a hollow tube maximizes bending stiffness per unit material and can raise critical speed. Because the force is independent of axial length, a stack of short, small-radius tubular modules can retain the field advantage while avoiding the severe bending-critical-speed penalty of one unsegmented long tube.

Central result

The field law rewards angular phase rate, total active mass, complete circumferential occupancy, and small mechanically sustainable radius. The mechanically preferred implementation is generally a balanced small-radius rotor or a coaxial stack of short tubular rotors, not an indefinitely long unsupported tube.

Contents

Section	Role
1. Central idea and scope	Core chapter
2. Coordinate lock and authoritative inputs	Core chapter
3. Native Cell3 geometry and field planes	Foundational geometry
4. Rotating tube geometry and mass measure	Core chapter
5. Elemental force law	Core derivation
6. Exact circumferential integration	Core derivation
7. Antipodal addition and cancellation	Core derivation
8. Steady force and zero torque	Core result
9. Calibrated reference rotor	Quantitative result
10. Comparison with point-mass devices	Device comparison
11. What efficiency means	Engineering interpretation
12. Fixed-angular-speed scaling	Scaling law
13. Fixed-surface-speed scaling	Scaling law
14. Axial mass packing and the long-tube limit	Scaling law
15. Hollow tube versus solid rod	Cross-section optimization
16. Distributed loading and rotor dynamics	Structural mechanics
17. Practical optimum and modular architecture	Design synthesis
18. Experimental predictions	Qualification plan
19. Conclusion	Core conclusion
Appendix A. Complete input ledger	Reference data
Appendix B. All 27 Cell3 nodes	Reference data
Appendix C. Scaling ledgers	Reference data
Appendix D. Minimal machine reconstruction kernel	Implementation reference
Appendix E. Glossary	Educational reference

1. Central idea and scope

The analyzed device is a body of rotating mass whose angular-velocity axis is aligned with the Earth-CMB traversal axis. The reference rate is $\omega=1$ rad/s. The starting point is the calibrated continuous-ring result for a zero-thickness uniform rim. A tube is formed by repeating that ring continuously along the axis.

The field response is governed by the phase-weighted Earth velocity $\mathbf{g}=\mathbf{G}_{\text{phase}} \mathbf{V}_E$. For an angular-velocity vector $\mathbf{\Omega}=\omega \mathbf{n}$, every complete axisymmetric cross-section contributes the same force per unit mass. The compact vector law is:

$$\mathbf{F}_{\text{axisymmetric}} = (\kappa_F M / 2) \mathbf{\Omega} \text{ cross } (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$$

The force depends on total mass and angular phase rate. It does not depend directly on ring radius, tube length, wall thickness, or radial mass distribution. Those quantities control mechanical energy, stress, stiffness, critical speed, thermal behavior, and packaging.

This separation is the central educational point. Coupling efficiency and mechanical efficiency are different optimization problems. The K-Field integral determines the force available from mass and angular rate. Classical rotor mechanics determines how economically and safely that angular rate can be sustained.

Reference tubular rotor visualization: $L=1.000$ m, $R=0.100$ m, $M=1.000$ kg

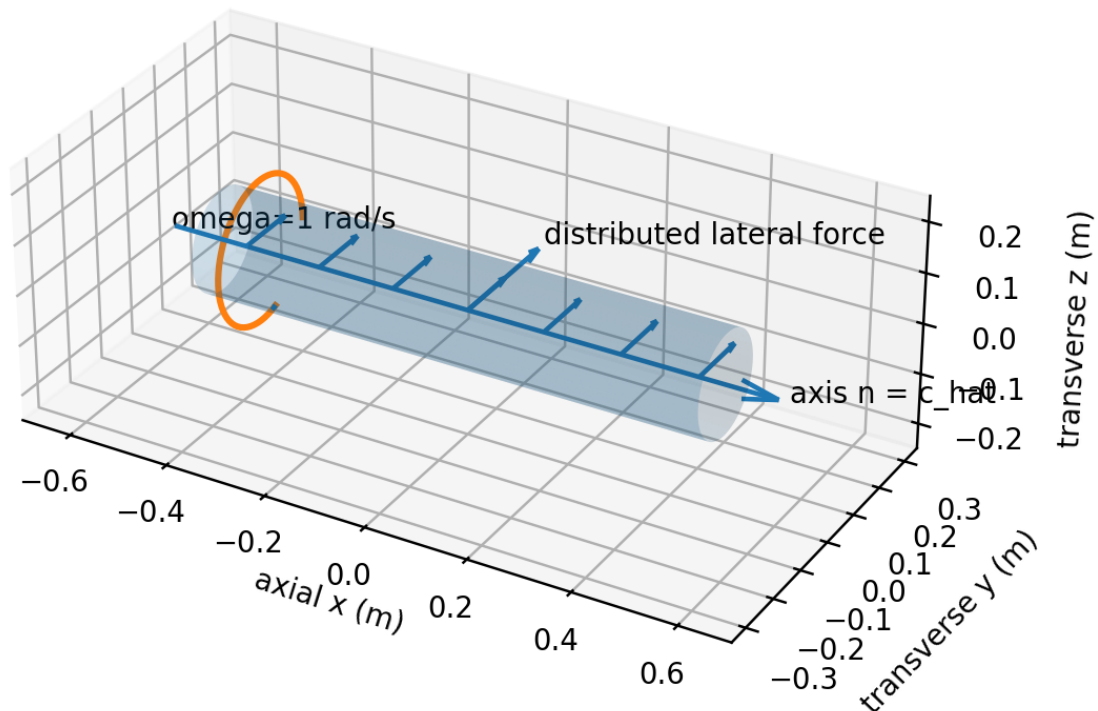


Figure 1. Dimensionally normalized tubular rotor. The displayed reference geometry is $L=1.000$ m and $R=0.100$ m with the actual reference mass $M=1.000$ kg and $\omega=1.000$ rad/s. The axial length is a visualization normalization; the field result depends only on total mass.

2. Coordinate lock and authoritative inputs

All vectors are expressed in ICRS/J2000 Cartesian coordinates. The origin is the rotor center. The unit CMB traversal axis $\hat{c}$, the two orthogonal rotor-plane basis vectors $\mathbf{e}_0$ and $\mathbf{A}_0$, and the orientation identity $\mathbf{e}_0 \times \mathbf{A}_0 = \hat{c}$ remove all phase and handedness ambiguity.

The Earth velocity $\mathbf{V}_E$ is first acted on by the symmetric physical response matrix $\mathbf{G}_{\text{phase}}$. This creates the effective phase-weighted velocity $\mathbf{g}$. Only the component of $\mathbf{g}$ perpendicular to the rotor axis contributes because a cross product removes the parallel component.

Quantity	Value	Dimension
$\hat{c}$	(-0.9707708558359, 0.207348009111, -0.120874929482)	unit vector
$\hat{c}$ RA, Dec	(167.943299880, -6.942599922)	deg
$\mathbf{e}_0$	(-0.2008648841096, -0.4262314799501, 0.8820317589697)	unit vector
$\mathbf{A}_0$	(0.1313668291131, 0.8805302542317, 0.4554220323958)	unit vector
$\mathbf{V}_E$	(-3.590031990658e+05, 7.667988603400e+04, -4.470106010069e+04)	m/s
$ \mathbf{V}_E $	3.698125020004e+05	m/s
$\mathbf{g} = \mathbf{G}_{\text{phase}} \mathbf{V}_E$	(-8.802219816351e+05, -1.439114365616e+05, 2.663350269486e+05)	m/s
$ \mathbf{g} $	9.308252172704e+05	m/s
$\text{angle}(\hat{c}, \mathbf{g})$	31.640861271139	deg
κ_F	3.07405089970203e-11	calibrated coupling

Physical response matrix $\mathbf{G}_{\text{phase}}$

	1	2	3
1	2.9764106378000	1.2431507795000	-2.0803600354000
2	1.2431507795000	2.5922286635000	-2.3178839937000
3	-2.0803600354000	-2.3178839937000	6.7735709133000

Exact orientation geometry: $\mathbf{F}$ is perpendicular to the CMB traversal axis

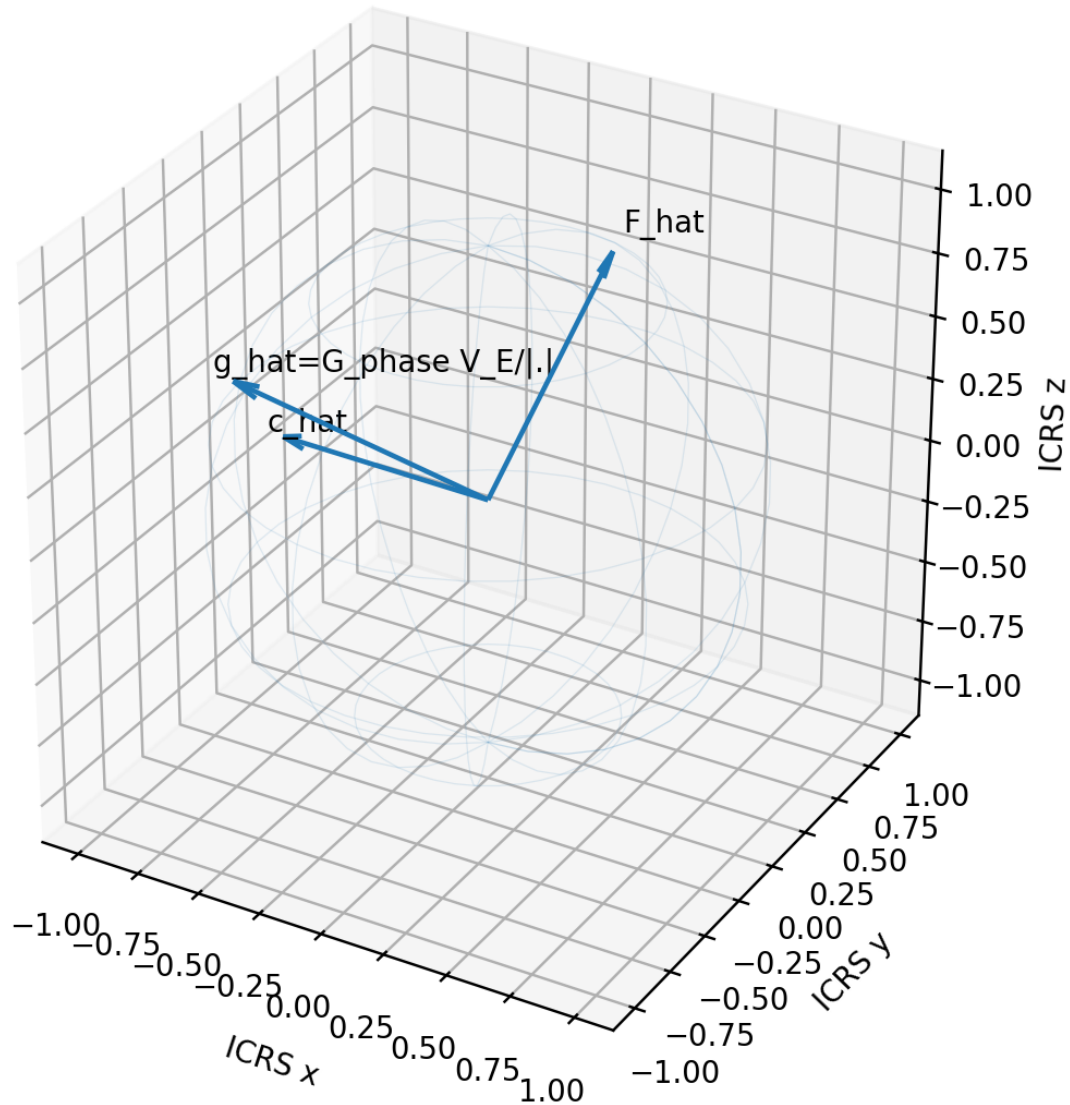


Figure 2. Exact ICRS vector geometry. The calibrated lateral force is parallel to $\mathbf{c_hat}$ cross $\mathbf{g}$ and is perpendicular to $\mathbf{c_hat}$ to numerical precision.

3. Native Cell3 geometry and field planes

The physical response matrix is part of the Cell3 field description. Cell3 is an anisotropic three-vector basis with unequal edge lengths, unequal included angles, three distinct face areas, and a nonzero signed volume. Its basis vectors establish the direct geometry from which the field response is constructed.

The centered 27-node cell uses integer labels $\mathbf{L}=(L1,L2,L3)$, with each coordinate taking values -1, 0, or +1. Physical coordinates are $p_L = B L/2$. This produces eight corners, twelve edge centers, six face centers, and one body center. The complete node set is antipodally symmetric.

Direct basis matrix B, columns a, b, c

Vector	ICRS x	ICRS y	ICRS z	Length
a	0.0203814385169464	-0.00341641449641055	-0.00233296591672896	0.0207970587824329
b	0.00219874596832336	0.0129021702586925	-0.0340665015360557	0.0364942051307272
c	-0.000630379485347372	-0.0242577220171279	-0.0429243545710938	0.0493085658998594

Primitive geometric ledger

Quantity	Included angle (deg)	Area / volume	ICRS unit normal
A_ab	83.933483012313	0.000754721789672624	(0.194092200808, 0.913177168821, 0.358379234791)
A_ac	80.448130018338	0.00101125575768807	(0.0890525941016, 0.866576769291, -0.491033948326)
A_bc	50.363231813756	0.00138578443173329	(-0.995965190309, 0.0836021625074, -0.0326192905644)
Cell volume	-	2.84206693780965e-05	native^3

Centered Cell3 base cell and all 27 address nodes

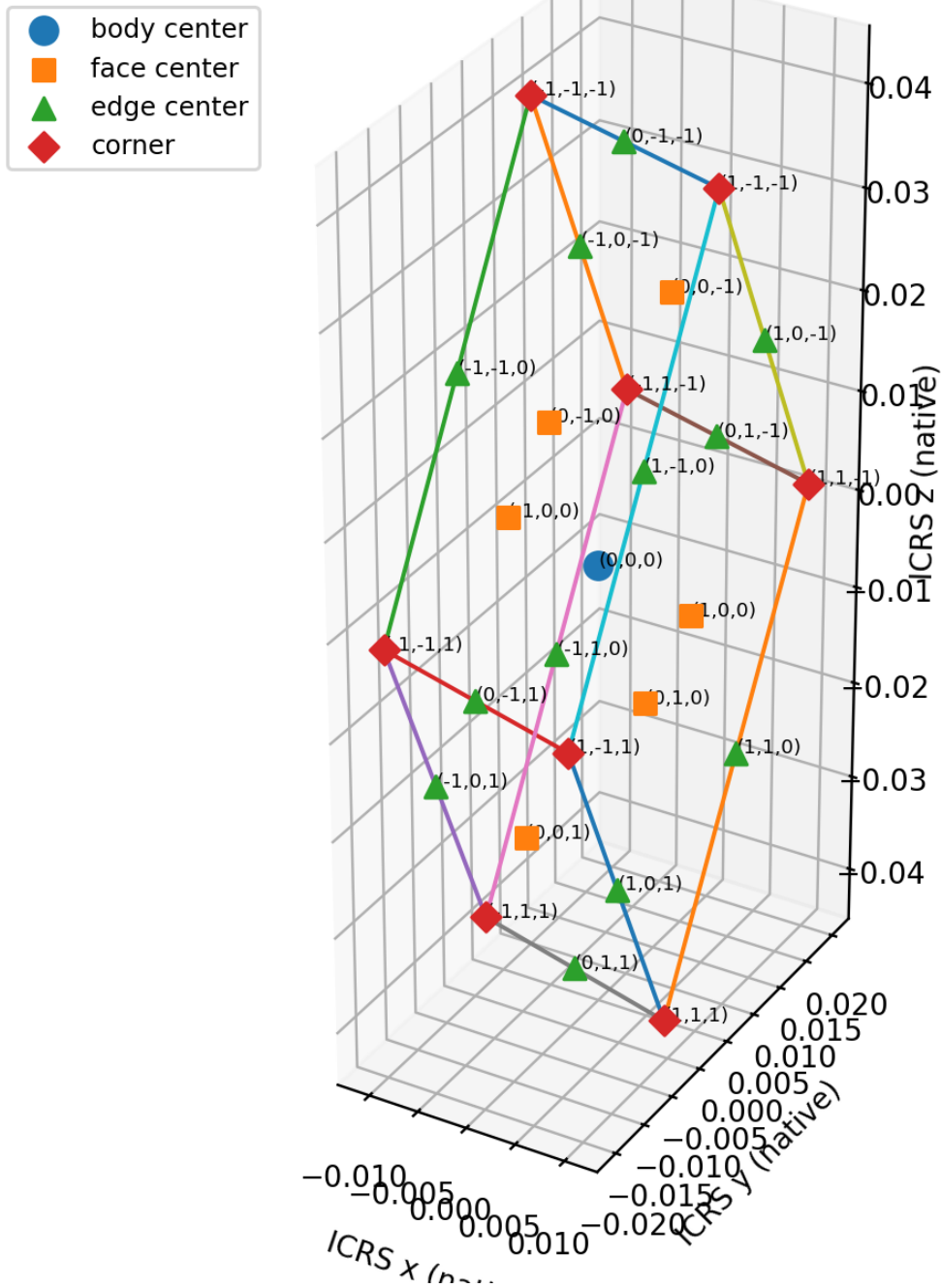


Figure 3. Dimensionally correct centered Cell3 in ICRS coordinates. Every one of the 27 address nodes is plotted and labeled. The colored node classes identify body center, face centers, edge centers, and corners.

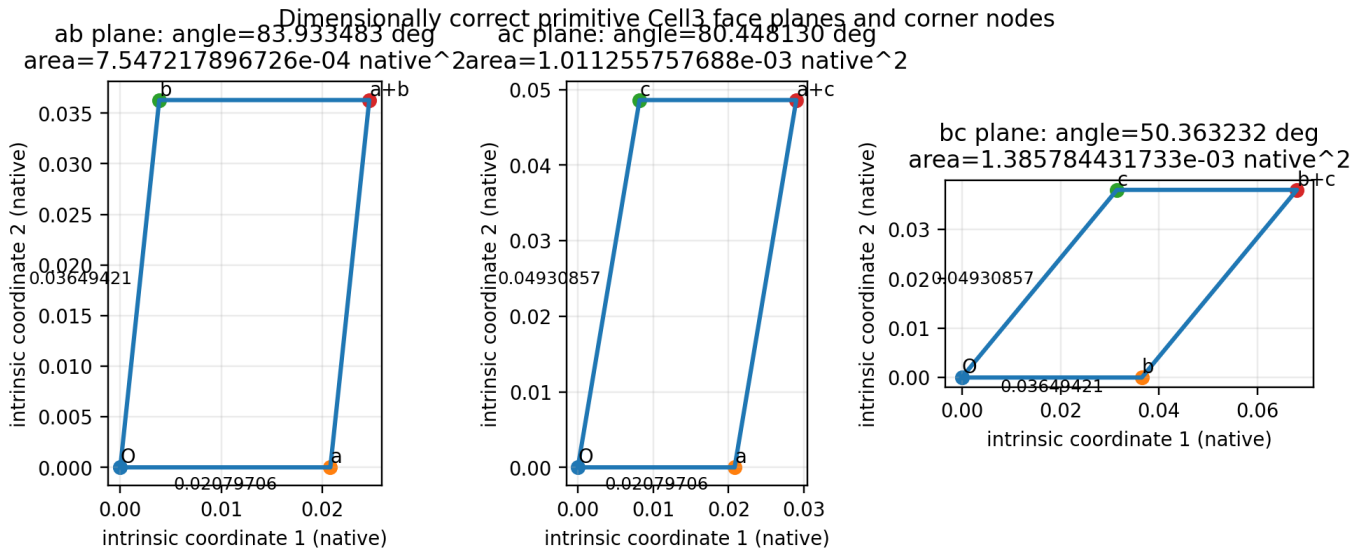


Figure 4. Dimensionally correct intrinsic views of the ab, ac, and bc primitive face planes. Each panel displays all four corner nodes, both edge lengths, the measured included angle, and the exact face area.

The largest face is the bc plane. Its area is 1.385784431733287e-3 native² and its included angle is 50.363231813756 deg. The three face normals are not interchangeable; their unequal orientations encode the anisotropy that enters G_{phase} . The rotor calculation therefore does not use an isotropic scalar field. It uses a fixed matrix acting on the Earth-CMB velocity before the rotational integration is performed.

4. Rotating tube geometry and mass measure

Let the tube have radius R , length L , total mass M , and axis n . At axial coordinate z and rim angle θ , the radial and tangential unit vectors are $u(\theta)$ and $t(\theta)$. They are identical for every axial slice. A uniform thin-walled tube has mass measure

$$dm = [M / (2 \pi L)] d\theta dz$$

Integrating dm from $z=-L/2$ to $+L/2$ and $\theta=0$ to 2π returns M exactly. The axial coordinate therefore appears only in the mass measure. Since the local cross-sectional geometry is the same at every z , the axial integral multiplies the ring result by total mass and introduces no new angular dependence.

For the locked rotor plane:

$$\begin{aligned} u(\theta) &= \cos(\theta)e_0 + \sin(\theta)A_0 \\ t(\theta) &= -\sin(\theta)e_0 + \cos(\theta)A_0 = c_{\text{hat}} \text{ cross } u(\theta) \\ r(\theta, z) &= z c_{\text{hat}} + R u(\theta) \\ v(\theta) &= V_E + R \omega t(\theta) \end{aligned}$$

5. Elemental force law

The complete elemental law contains a translation cross-term and a rotational self-term. Define $B(\theta) = t(\theta)^T G_{\text{phase}}$ and $C(\theta) = t(\theta)^T G_{\text{phase}} t(\theta)$. Then

$$dF = -\kappa_F dm [\omega B(\theta) + R \omega^2 C(\theta)] u(\theta)$$

The first term couples the rotating tangential direction to the phase-weighted Earth velocity. It is linear in ω . The second term is quadratic in ω and depends only on the local rotating velocity. The continuous-ring result is controlled

by how these two terms transform under $\theta \rightarrow \theta + \pi$.

The simplified source expression used for the lateral-force derivation is the translation term:

$$dF_{\text{translation}} = -\kappa_F \omega [t(\theta)^T G_{\text{phase}} V_E] u(\theta) dm$$

This expression is an integration of the supplied source geometry and coupling law. The source defines G_{phase} , the Earth-CMB velocity, and the force dependence on the velocity cross-term. No empirical curve fit is inserted during the rim integration.

6. Exact circumferential integration

The tube integral separates into axial and angular factors. The axial factor is total mass. The angular factor is a second-moment identity in the rotor plane:

$$\int_0^{2\pi} u(\theta) t(\theta)^T d\theta = \pi [e_0 A_0^T - A_0 e_0^T]$$

The bracketed matrix is the antisymmetric cross-product operator associated with the rotor normal. Applying it to $g = G_{\text{phase}} V_E$ gives the exact ring and tube result:

$$F_{\text{tube}} = (\kappa_F M \omega / 2) \hat{c} \text{ cross } (G_{\text{phase}} V_E)$$

$$F_{\text{tube}} = (\kappa_F M / 2) \Omega \text{ cross } (G_{\text{phase}} V_E), \text{ where } \Omega = \omega \hat{c}$$

Neither R nor L appears. At a fixed angular speed, a ring, thin cylindrical shell, thick annulus, solid cylinder, or axial stack of complete rings produces the same ideal force per unit active mass. The result requires complete axisymmetric phase occupancy. An incomplete sector retains angular modulation.

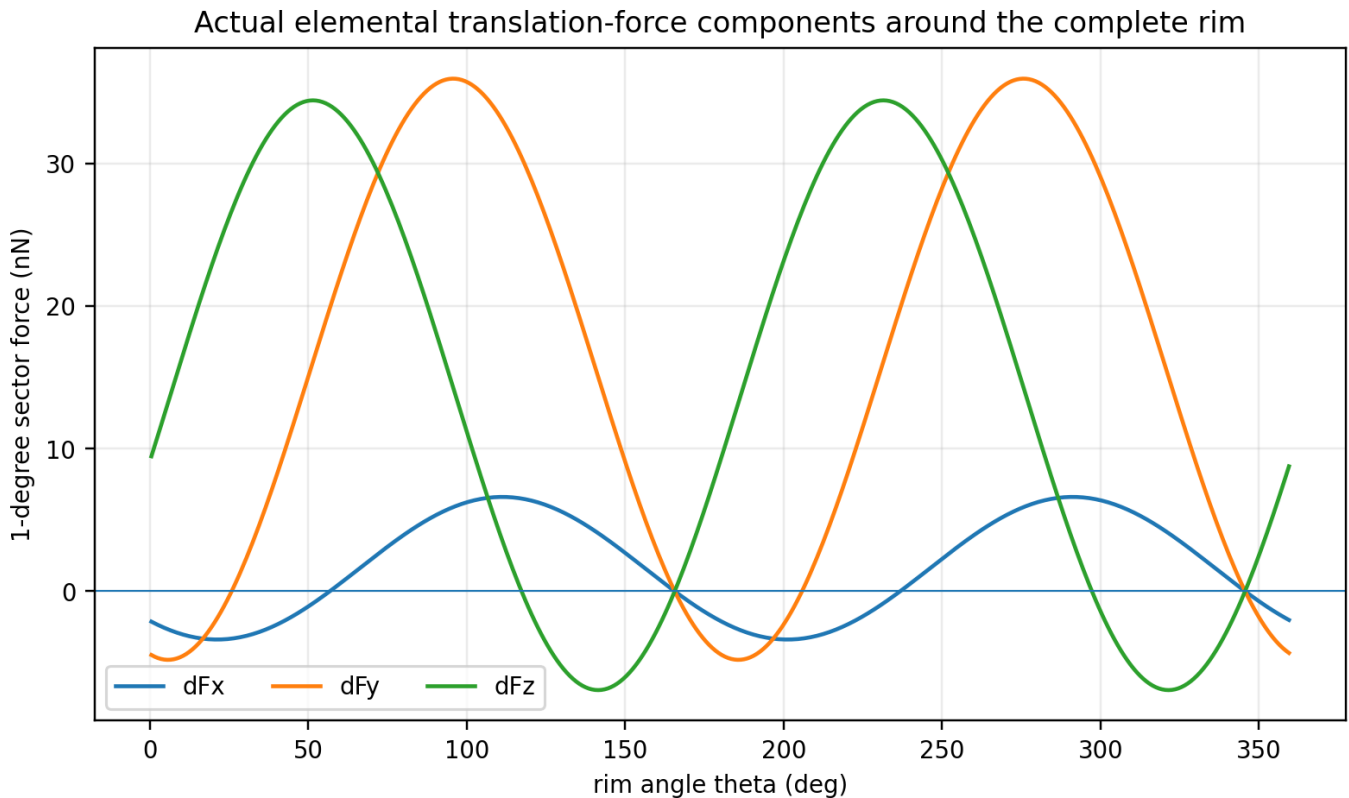


Figure 5. Actual 1-degree elemental translation-force components around the reference ring. Each component oscillates with angle, but the complete vector integral is nonzero.

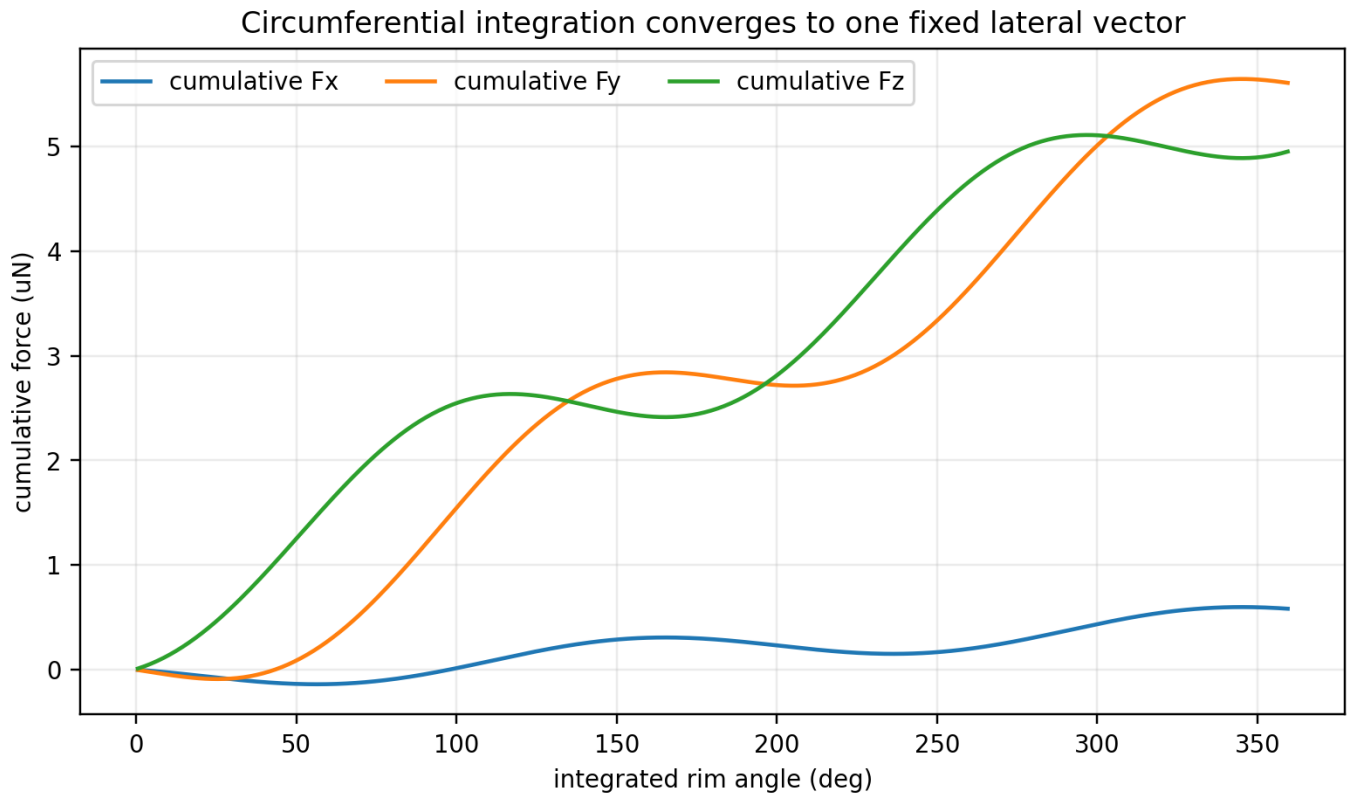


Figure 6. Cumulative 1-degree integration. The angularly varying elements converge to the fixed calibrated vector (0.581438, 5.609329, 4.952562) uN.

7. Antipodal addition and cancellation

Every point on the rim has an opposite partner. For that partner, $u(\theta + \pi) = -u(\theta)$ and $t(\theta + \pi) = -t(\theta)$. The translation kernel B therefore changes sign, while the radial vector also changes sign. The two sign reversals cancel:

$$[t(\theta + \pi)^T G_{\text{phase}} V_E] u(\theta + \pi) = [t(\theta)^T G_{\text{phase}} V_E] u(\theta)$$

Opposite translation-force elements point in the same direction and add. The remaining angular phases fill the complete circle. Oscillating components cancel, while the average vector survives.

The rotational self-kernel $C(\theta) = t^T G_{\text{phase}} t$ is unchanged under the antipodal transformation because t changes sign twice inside the quadratic form. Multiplication by u then reverses the self-force. Therefore the quadratic self-term cancels exactly between antipodal partners:

$$dF_{\text{self}}(\theta + \pi) = -dF_{\text{self}}(\theta)$$

This leaves the linear translation cross-term as the complete net force of a uniform ring or tube. The actual 1-degree source ledger gives self-force elements near 10^{-14} N at the reference rate; their complete sum is zero to numerical quadrature precision.

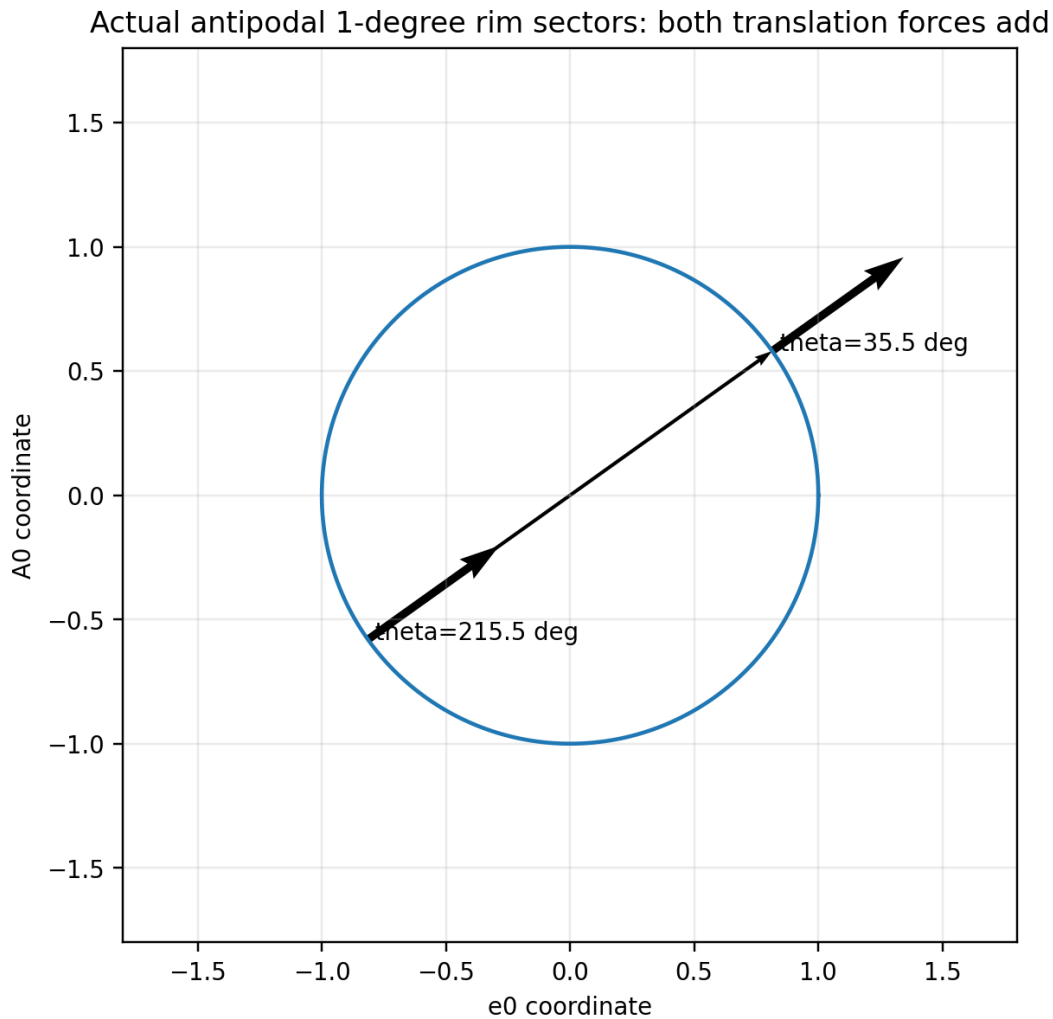


Figure 7. Actual antipodal 1-degree sectors at $\theta=35.5$ deg and 215.5 deg. Radial and tangential directions reverse together, so the translation-force arrows point in the same rotor-plane direction and add.

8. Steady force and zero torque

A complete uniform ring places mass at every dumbbell phase simultaneously. It converts the pulsating point-pair force into its constant cycle-average value. The net vector does not rotate with the body; it remains fixed in the ICRS direction set by $\mathbf{c}_{\text{hat}} \times \mathbf{g}$.

For each cross-sectional element, the elemental force is radial, so $\mathbf{r}_{\text{perp}} \times d\mathbf{F} = 0$ about the local slice center. For a centered tube, the axial moment $\mathbf{z}_{\text{hat}} \times d\mathbf{F}$ cancels between $+z$ and $-z$ slices. The total torque about the tube center is therefore zero:

$$\tau_{\text{tube}} = 0 \text{ for a centered, axially symmetric mass distribution}$$

The absence of net torque does not mean the material is unloaded. Each material sector rotates through the fixed inertial force pattern and experiences a cyclic local body force in material coordinates. Bearings and external supports see the steady integrated lateral load, while the rotating shell sees a periodic stress history. This distinction is important for fatigue and whirl analysis.

9. Calibrated reference rotor

The reference device has $M=1$ kg, $R=0.10$ m, and $\omega=1$ rad/s. The axis is $\hat{c}$. The calibrated result is

$$\mathbf{F}_{\text{tube}} = (0.5814375586, 5.6093290679, 4.9525619648) \text{ uN}$$

$$|\mathbf{F}_{\text{tube}}| = 7.5053655635 \text{ uN}$$

The coupling interval gives $7.4289192 \text{ uN} \leq |\mathbf{F}_{\text{tube}}| \leq 7.5818120 \text{ uN}$. The direction is $RA=84.0821143$ deg and $Dec=41.2899223$ deg. The dot product with $\hat{c}$ is $-4.789\text{e-}23$ N, so the force is perpendicular to the CMB axis to numerical precision.

The reference kinematic quantities are:

Quantity	Value	Dimension
Mass M	1.000000000	kg
Radius R	0.100000000	m
Angular speed ω	1.000000000	rad/s
Rotation rate	9.549296585514	rpm
Period	6.283185307180	s
Surface speed	0.100000000000	m/s
Moment of inertia, thin shell	0.010000000000	kg m ²
Rotational energy	0.005000000000	J
Angular momentum	0.010000000000	kg m ² /s
Ideal hoop tension	0.015915494309	N
Impulse per revolution	4.715760263345e-05	N s

The impulse per revolution is independent of angular speed under the linear- ω law because F is proportional to ω while the period is proportional to $1/\omega$:

$$J_{\text{revolution}} = F (2 \pi / \omega) = \pi \kappa_F M |\hat{c} \text{ cross } \mathbf{g}|$$

10. Comparison with point-mass devices

A single-axis dumbbell with the same total moving mass of 1 kg has a peak lateral force of approximately 15.010731 uN and a cycle average of 7.505366 uN . Therefore the uniform ring or tube equals the dumbbell cycle average and one-half of its peak. The tube replaces time modulation with simultaneous phase occupancy.

The 1 kg ring also produces the same average lateral force as the earlier 2 kg dual-axis reference device at base rate $\omega_B=1$ rad/s, while eliminating the second rotational axis and using half the active mass. The dual-axis device retains much larger instantaneous modulation, but its cycle-average translation is the same calibrated value.

Device	Active mass (kg)	Peak/steady force (uN)	Cycle average (uN)	Character
Uniform ring, axis = $\hat{c}$	1.000	7.505366	7.505366	steady
Uniform ring, maximum pure-lateral axis	1.000	14.307020	14.307020	steady
Single-axis antipodal point pair, total mass 1 kg, axis = $\hat{c}$	1.000	15.010731	7.505366	pulsating
Dual-axis 2:1 reference, two 1 kg masses	2.000	127.068062	7.505366	strongly modulated

11. What efficiency means

At fixed angular speed, the tube is not intrinsically superior in force per active mass. The calibrated mass-specific law is

$$F/M = 7.5053655635 \text{ } \omega \text{ uN/kg, with } \omega \text{ in rad/s}$$

$$F/M = 0.7859600439 \text{ RPM uN/kg}$$

A ring, cylindrical shell, thick annulus, solid cylinder, and stack of rings all produce the same ideal force per unit rotating mass at the same ω . The long tube advantage is therefore mechanical and architectural: it supplies continuous circumferential phase occupancy, permits mass to be accumulated axially at small radius, and distributes the steady load along a manufacturable body.

Four efficiency measures must be kept separate. Coupling efficiency is F/M at a specified ω . Energy efficiency is force per stored rotational energy. Stress efficiency is force at a specified allowable material speed or stress. Package efficiency is force per unit length, volume, or bearing span. A geometry can improve one while degrading another.

Efficiency measure	Metric	Controlling result
Coupling efficiency	F/M	Depends on ω ; independent of R , L , and axisymmetric cross-section
Energy efficiency	F/E_{rot}	Improves as radius of gyration decreases at fixed ω
Surface-speed efficiency	F/M at fixed v_{max}	Improves as $1/R_{\text{outer}}$
Package-length efficiency	F/L	At fixed wall areal density and surface speed, independent of radius
Structural efficiency	critical speed and bending stiffness per mass	Favors tubular sections and short bearing spans

12. Fixed-angular-speed scaling

At fixed angular speed, the force law is

$$F_{\text{tube}} = 7.5053655635 (M/1 \text{ kg}) (\omega/1 \text{ rad/s}) \text{ uN}$$

The 10 cm reference radius does not affect the predicted K-Field force at fixed ω . It controls surface speed, moment of inertia, stored energy, angular momentum, and centrifugal loading. For a thin shell:

$$v_{\text{surface}} = R \omega; I = M R^2; E_{\text{rot}} = (1/2) M R^2 \omega^2$$

$$\text{ideal hoop tension } T = M R \omega^2 / (2 \pi)$$

At $\omega = 1 \text{ rad/s}$, reducing radius by a factor of ten leaves force unchanged, reduces thin-shell rotational energy by a factor of one hundred, and reduces the reference ideal hoop tension by a factor of ten. Material hoop stress scales with $R^2 \omega^2$ when expressed as ρv^2 .

R (m)	F (uN)	surface speed (m/s)	I shell (kg m ²)	E_rot (J)	ideal T (N)
0.100000	7.5053656	0.100000000	1.0000000000000e-02	5.000000000000e-03	1.591549430919e-02
0.010000	7.5053656	0.010000000	1.000000000000e-04	5.000000000000e-05	1.591549430919e-03
0.001000	7.5053656	0.001000000	1.000000000000e-06	5.000000000000e-07	1.591549430919e-04

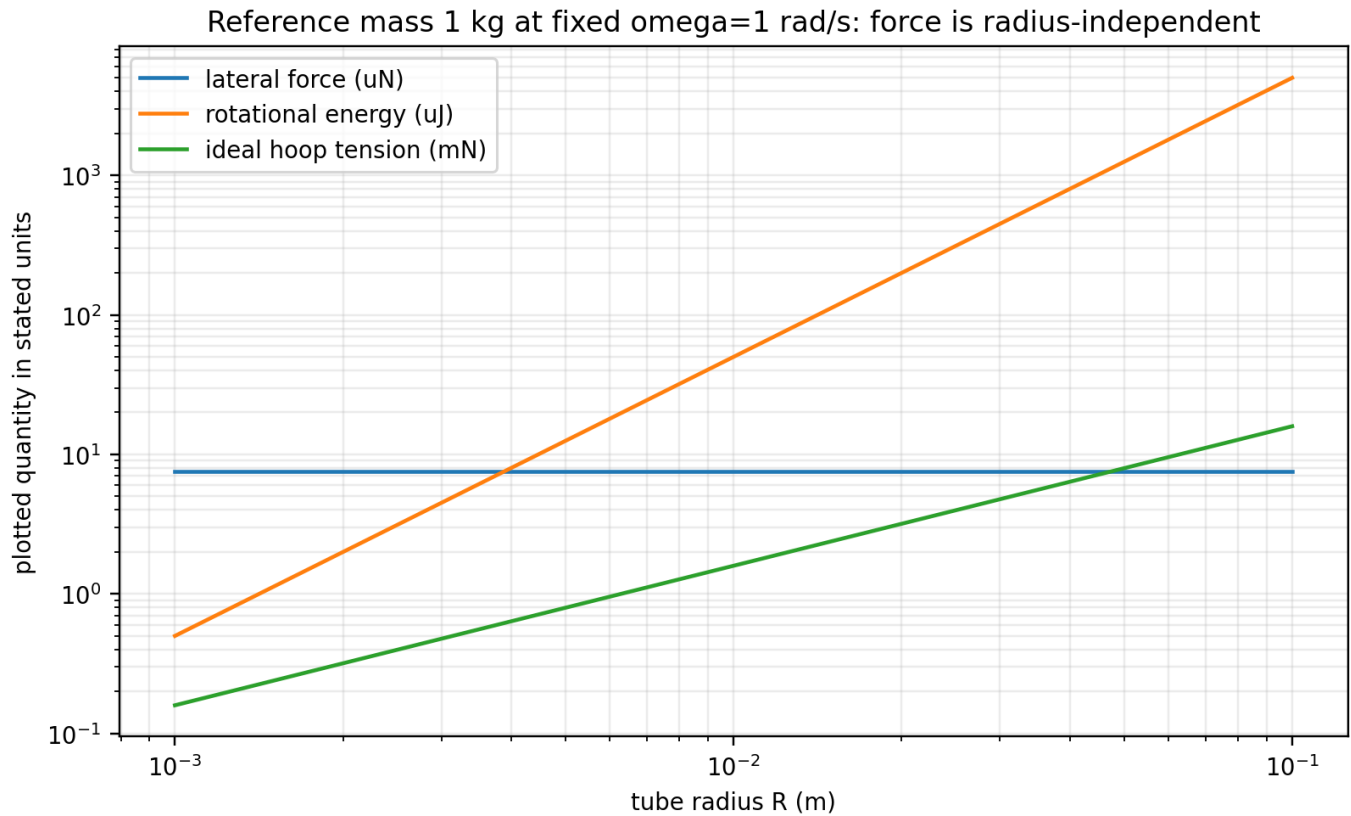


Figure 8. Exact fixed-omega scaling for the 1 kg thin-shell reference. The force is flat with radius; energy falls as R^2 and the ideal tension scale falls as R .

13. Fixed-surface-speed scaling

The stronger small-radius result appears when the mechanical constraint is surface speed or a stress limit rather than angular speed. Since $\omega=v/R$,

$$F/M = K v/R, \text{ where } K=7.5053655635e-6 \text{ m/s}$$

At fixed surface speed, force per mass increases as $1/R$. For a thin shell, rotational energy per mass remains $(1/2)v^2$, and the dominant hoop-stress scale remains proportional to ρv^2 . Smaller radius therefore increases angular phase rate without increasing surface speed.

Using the reference surface speed $v=0.10$ m/s:

R (m)	ω (rad/s)	rpm	F for 1 kg (uN)	E_rot shell (J)
0.100000	1.000000	9.549297	7.505366	0.005000000
0.010000	10.000000	95.492966	75.053656	0.005000000
0.001000	100.000000	954.929659	750.536556	0.005000000

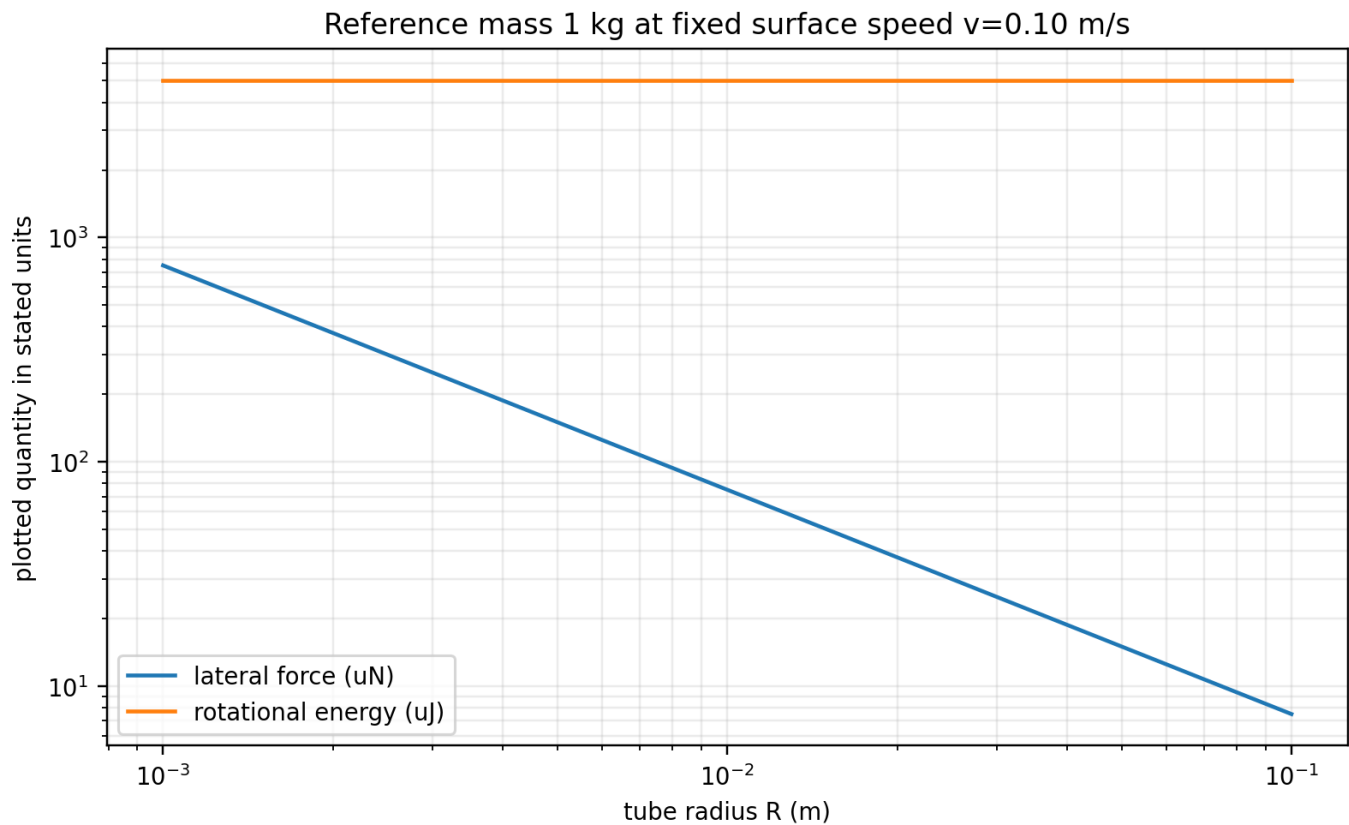


Figure 9. Exact fixed-surface-speed scaling. Force rises as $1/R$ while thin-shell rotational energy remains constant.

This is the principal reason the long, narrow rotor concept emerges. Reducing diameter permits a proportionally higher angular phase rate without raising surface speed. The field gain is not created by tube length; length is used to retain or accumulate active mass after radius is reduced.

14. Axial mass packing and the long-tube limit

For a thin tube with wall thickness t and material density ρ ,

$$M = 2 \pi R L \rho t$$

At fixed mass, material, and wall thickness, L is proportional to $1/R$. Reducing radius by a factor of ten requires ten times the axial length to retain the same active mass. The geometry exchanges radial envelope for axial length.

At fixed surface speed, substituting $\omega=v/R$ gives

$$F = 2 \pi K \rho t L v$$

$$F/L = 2 \pi K \rho t v$$

Force per unit tube length is independent of radius when wall areal density ρt and surface speed are fixed. A narrower tube increases force per unit mass because each kilogram rotates through phase faster, but the same construction contains less mass per unit length. The required length rises in the compensating ratio.

This establishes a precise limit on the phrase most efficient. A narrow tube improves force per mass, force per stored energy at fixed ω , and radial-envelope efficiency. It does not automatically improve force per package length.

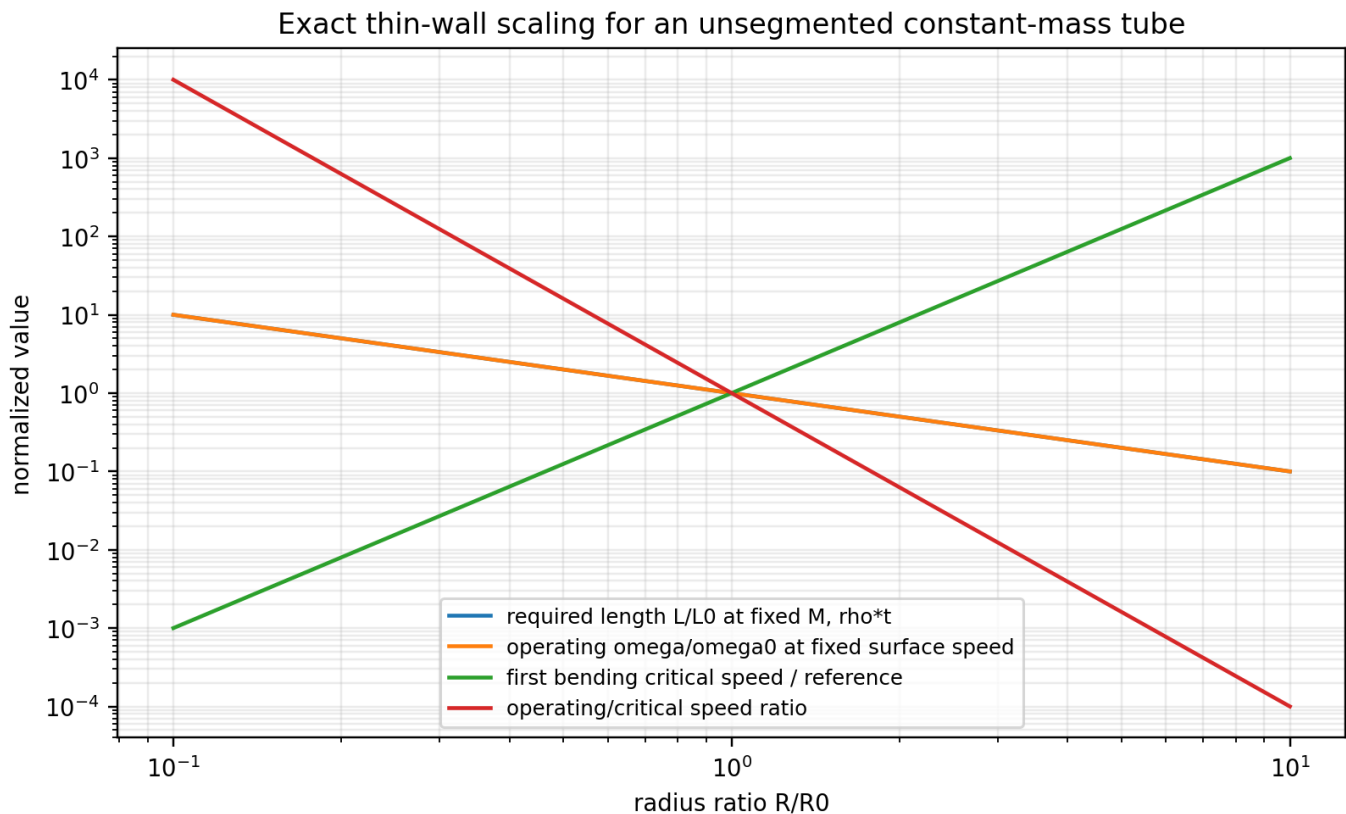


Figure 10. Normalized exact thin-wall power laws for one unsegmented constant-mass tube. The field-preferred operating speed rises as $1/R$ at fixed surface speed, but the first bending critical speed falls as R^3 when the required unsupported length grows as $1/R$.

15. Hollow tube versus solid rod

The field integral depends on mass, not on the radius at which that mass resides. This means the K-Field law alone does not distinguish a hollow tube from a solid rod at the same M and ω . Classical mechanics does.

For any axisymmetric cross-section, define the mass radius of gyration k_r by $I_p = M k_r^2$. The stored rotational energy is

$$E_{\text{rot}} = (1/2) M k_r^2 \omega^2$$

A thin cylindrical shell has $k_r = R$. A uniform solid cylinder has $k_r = R/\sqrt{2}$. At the same mass, outer radius, and angular speed, the solid cylinder stores one-half the rotational energy of the thin shell while producing the same K-Field force. Therefore a solid rod is superior when minimum spin energy is the only mechanical objective.

A hollow tube is superior in another dimension. Material placed away from the neutral axis produces much greater transverse area moment of inertia and bending stiffness per unit mass. That raises resistance to sag, imbalance deflection, and bending modes. The tube therefore exchanges higher polar spin inertia for higher beam stiffness. The optimal cross-section is selected by the dominant constraint: energy favors central mass; bending critical speed favors material at larger radius.

Field coupling depends on mass; stored spin energy depends on radius of gyration

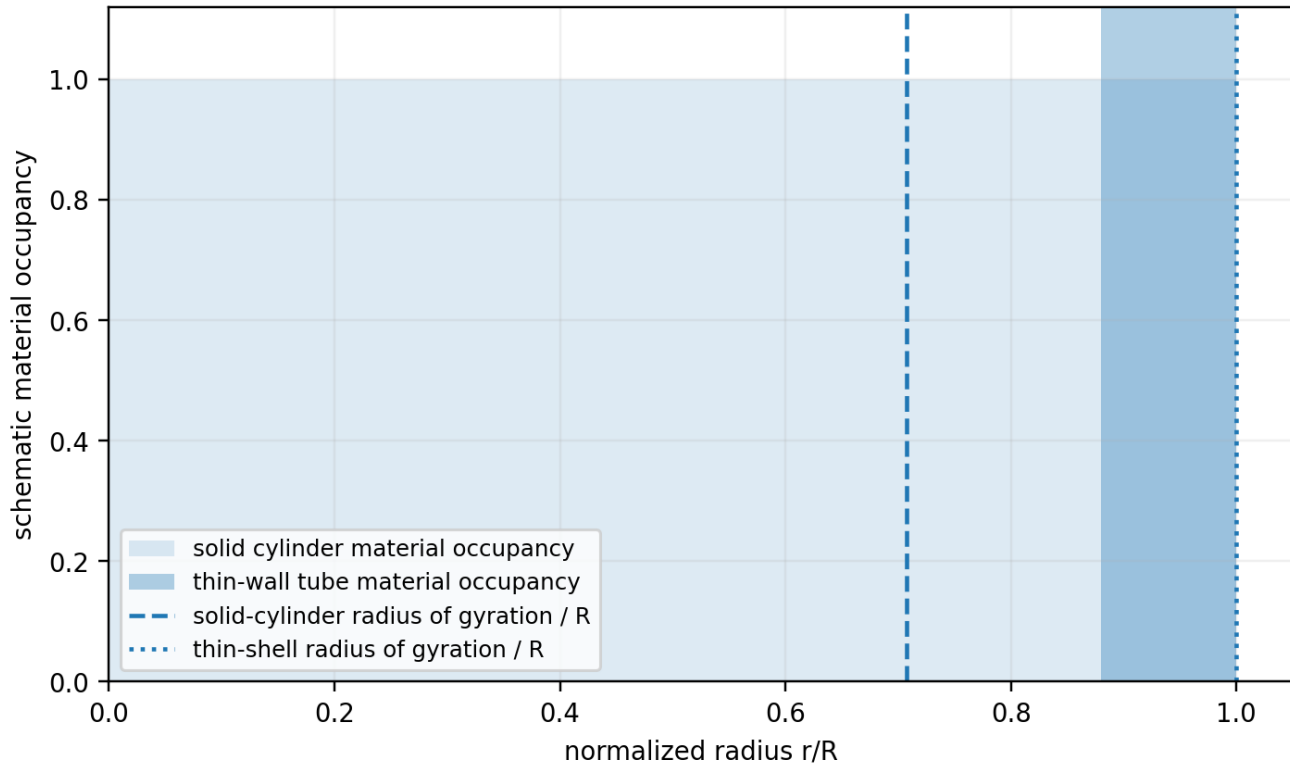


Figure 11. Cross-sectional optimization. Field force follows total mass. Rotational energy follows mass radius of gyration. Bending stiffness follows the transverse area moment and therefore strongly rewards tubular placement of material.

The strongest design conclusion is consequently not hollow tube under every condition. It is small-radius, axisymmetric, fully occupied rotating mass, with the radial distribution chosen to satisfy the structural constraint. The hollow tube is especially valuable when the rotor must be long enough that bending stiffness and critical speed dominate.

16. Distributed loading and rotor dynamics

A uniform tube generates the same force per unit axial mass. If the tube is freely translating, every slice shares the acceleration $a_K = F/M$ and no static bending is required. If bearings or a test frame restrain translation, the K-Field force becomes a distributed transverse load $q_K = F/L$.

For a simply supported uniform span, the classical screening relations are

$$\begin{aligned} \text{reaction at each support} &= F/2 \\ \text{maximum bending moment} &= q_K L^2/8 = F L/8 \\ \text{maximum static deflection} &= 5 q_K L^4/(384 E I_{\text{area}}) \end{aligned}$$

Supported tube: steady net force becomes a distributed transverse beam load

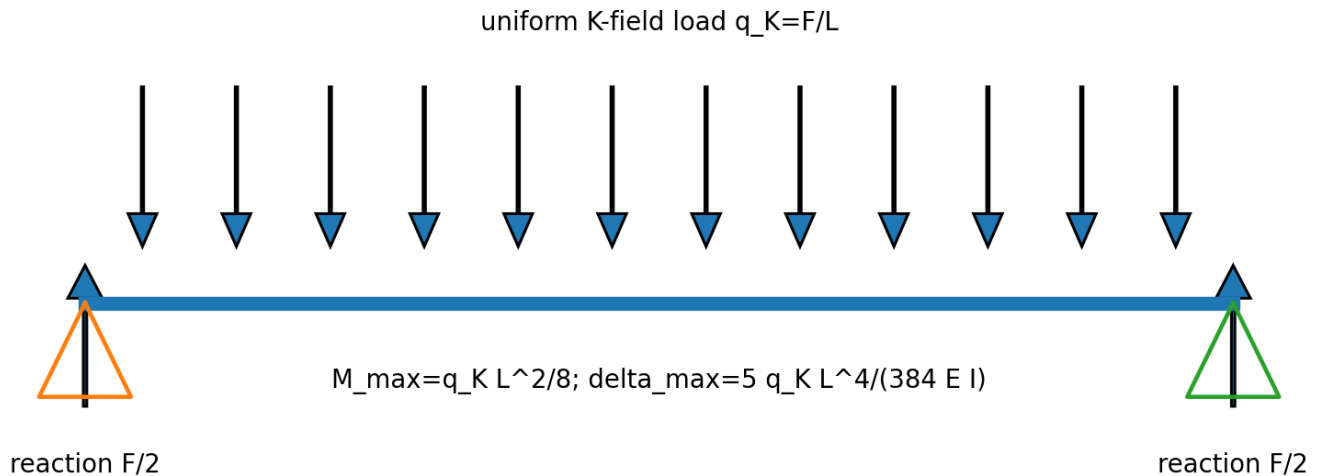


Figure 12. Supported tube under the steady integrated lateral load. The field result is distributed by axial mass, so support reactions and beam bending grow with span length.

The first simply supported bending frequency has the scaling

$$\omega_{\text{crit},1} = \pi^2 \sqrt{E I_{\text{area}} / (\mu L^4)}$$

For a thin-wall circular tube, I_{area} is approximately $\pi R^3 t$ and mass per length μ is approximately $2 \pi R \rho t$. Thus ω_{crit} is proportional to R/L^2 . If total mass, wall thickness, and material are fixed, L is proportional to $1/R$, so ω_{crit} is proportional to R^3 .

At fixed allowable surface speed, the desired operating ω is proportional to $1/R$. The ratio of operating speed to first bending critical speed therefore grows as $1/R^4$ for one unsegmented tube. This is the dominant mechanical counterweight to the field-side small-radius gain. The mathematical force law drives R downward; unsegmented rotor dynamics drives the design toward a finite radius and finite bearing span.

17. Practical optimum and modular architecture

The ideal architecture has the smallest mechanically sustainable radius, the largest feasible active-mass fraction, complete circumferential symmetry, and the highest sustainable angular speed. It also requires adequate bending stiffness, short unsupported spans, precise balance, low bearing loss, thermal control, and a support structure capable of measuring a steady lateral load.

Because the field force is independent of axial length, one long tube is not required. A coaxial stack of short tubes or short solid rotors produces the same total force when total mass and angular speed are unchanged. Axial segmentation can preserve the field advantage while raising bending critical speed, reducing sag, simplifying balancing, and isolating rotor failures.

A practical hierarchy follows. First select the angular-velocity vector and required force. Next choose the smallest radius compatible with surface speed, manufacturing tolerance, and bearing technology. Then choose the cross-section by trading spin energy against bending stiffness. Finally divide the required mass into enough supported modules that no rotor operates near a bending critical speed.

Design synthesis

The field-preferred object is a small-radius axisymmetric rotor. The mechanically preferred embodiment is often a modular coaxial array of short tubular rotors. A single very long narrow tube is useful only when its bending stiffness and bearing support keep the complete operating range clear of critical speeds.

The CMB-aligned reference is not the maximum-amplitude orientation. Holding the requirement of pure lateral force while optimizing the rotor axis produces 14.307020 uN for 1 kg at 1 rad/s, a gain of 1.906239 over the CMB-aligned value. That orientation is RA=116.050867 deg, Dec=-44.008761 deg and is tilted 58.359139 deg from $\hat{c}$. This orientation result is separate from the tube-radius scaling: every axisymmetric cross-section retains the same mass and angular-speed laws.

18. Experimental predictions

The tubular-rotor result produces a compact set of discriminating predictions.

1. At fixed M, ω , and axis orientation, force is independent of tube radius within the linear- ω regime.
2. At fixed M and allowable surface speed, force varies as $1/R$.
3. At fixed M and ω , changing tube length does not change force.
4. Dividing one tube into coaxial short modules does not change total force when total active mass and angular velocity are preserved.
5. A complete uniform circumference produces a steady inertial force. Removing a sector introduces periodic modulation.
6. Reversing the angular-velocity vector reverses the force vector.
7. The calibrated CMB-aligned force direction remains RA=84.082114 deg, Dec=41.289922 deg and is perpendicular to $\hat{c}$.
8. A centered rotor produces no net K-Field torque, although local material stress cycles with rotor phase.
9. The impulse per revolution remains constant while force grows linearly with ω .
10. At sufficiently small radius or sufficiently high ω , the linear law must enter a transition. Otherwise F/M diverges as R approaches zero at fixed surface speed.

Recommended qualification sequence

Test	Controlled change	Predicted result
A. Radius invariance	Use equal mass rotors at several radii and identical ω	F constant within calibration and measurement uncertainty
B. Angular-rate linearity	Sweep ω at fixed rotor	F proportional to ω ; fixed inertial direction
C. Spin reversal	Reverse Ω	F reverses
D. Circumferential completeness	Compare full ring with removable sectors	Full ring steady; sectors introduce harmonics
E. Axial segmentation	Compare one tube with equal-mass short rotor stack	Same total mean force
F. Fixed-speed radius sweep	Hold surface speed and mass constant	F proportional to $1/R$ until transition
G. Small-radius transition	Extend the fixed-speed sweep	Locate coherence, saturation, or nonlinear response scale

The small-radius sweep is the decisive model-completion experiment. The present linear- ω law predicts unbounded force per mass as R approaches zero at fixed surface speed. A physical K-Field description therefore contains a transition scale. The radius, angular rate, and spectral character at that transition directly measure previously unresolved field

properties such as coherence length, saturation response, or rate-dependent coupling.

19. Conclusion

The exact tube integral is identical to the exact ring integral because axial integration contributes only total mass. A complete uniform tube therefore produces a steady lateral force, not a rotating force pulse. At $M=1$ kg and $\omega=1$ rad/s with the axis aligned to the CMB traversal direction, the calibrated force is 7.505365563 μN in the fixed ICRS direction $RA=84.082114$ deg, $Dec=41.289922$ deg.

At fixed ω , the long narrow tube does not increase force per mass. It reduces mechanical cost by allowing small radius and axial mass packing. At fixed surface speed, its force per mass grows as $1/R$. The price of retaining mass is length proportional to $1/R$, and the price of leaving that length unsupported is a bending critical speed proportional to R^3 . The full optimization therefore has a finite mechanical optimum.

The field law selects complete circumferential occupancy, small radius, high angular phase rate, and total active mass. Structural mechanics selects the cross-section and bearing span. A solid rod minimizes spin energy; a hollow tube maximizes bending stiffness per mass; a modular stack of short small-radius rotors avoids the critical-speed penalty of one long tube. This combined field-and-mechanics result identifies the modular narrow tubular rotor as the strongest general actuator architecture under the current calibrated law.

The remaining boundary is the high- ω , small-radius transition. Measuring that boundary converts the rotor from an actuator concept into a probe of K-Field coherence, rate dependence, and saturation.

Appendix A. Complete input ledger

A.1 Coordinate and field vectors

Input	Value	Dimension
c_hat x	-0.9707708558359351	unit
c_hat y	0.2073480091109861	unit
c_hat z	-0.1208749294819919	unit
e0 x	-0.2008648841095591	unit
e0 y	-0.4262314799500942	unit
e0 z	0.882031758969713	unit
A0 x	0.1313668291130585	unit
A0 y	0.880530254231655	unit
A0 z	0.4554220323957963	unit
V_E x	-3.590031990658130e+05	m/s
V_E y	7.667988603400199e+04	m/s
V_E z	-4.470106010068800e+04	m/s
g x	-8.802219816350914e+05	m/s
g y	-1.439114365615995e+05	m/s
g z	2.663350269486193e+05	m/s
kappa recommended	3.0740508997020345e-11	coupling
kappa lower	3.0427399447546500e-11	coupling
kappa upper	3.1053618546494197e-11	coupling

A.2 Direct metric $G=B^T B$

	1	2	3
1	4.325176539999714e-04	8.021043124662487e-05	1.701674486382120e-04
2	8.021043124662487e-05	1.331827008123595e-03	1.147919287025428e-03
3	1.701674486382120e-04	1.147919287025428e-03	2.431334671100779e-03

A.3 Exact force ledger

Quantity	Value	Dimension
Aligned force x	5.814375586393292e-07	N
Aligned force y	5.609329067853807e-06	N
Aligned force z	4.952561964816686e-06	N
Aligned magnitude	7.505365563476241e-06	N
Lower magnitude	7.428919151009735e-06	N
Upper magnitude	7.581811975942748e-06	N
Force RA	84.082114313919035	deg
Force Dec	41.289922334276525	deg
Impulse per revolution	4.715760263344556e-05	N s
Optimized pure-lateral magnitude	1.430702048307761e-05	N
Optimized gain	1.906238991568995	ratio

Appendix B. All 27 Cell3 nodes

Every node uses $p_L = B L/2$ with L_i in $\{-1, 0, +1\}$. Coordinates are ICRS native units. The ledger is complete and antipodally paired.

L1	L2	L3	ICRS x	ICRS y	ICRS z	Class
-1	-1	-1	-0.010974902499961	0.007385983127423	0.039661911011939	corner
-1	-1	0	-0.011290092242635	-0.004742877881141	0.018199733726392	edge center
-1	-1	1	-0.011605281985309	-0.016871738889705	-0.003262443559155	corner
-1	0	-1	-0.009875529515800	0.013837068256769	0.022628660243911	edge center
-1	0	0	-0.010190719258473	0.001708207248205	0.001166482958364	face center
-1	0	1	-0.010505909001147	-0.010420653760359	-0.020295694327182	edge center
-1	1	-1	-0.008776156531638	0.020288153386115	0.005595409475884	corner
-1	1	0	-0.009091346274312	0.008159292377552	-0.015866767809663	edge center
-1	1	1	-0.009406536016985	-0.003969568631012	-0.037328945095210	corner
0	-1	-1	-0.000784183241488	0.005677775879218	0.038495428053575	edge center
0	-1	0	-0.001099372984162	-0.006451085129346	0.017033250768028	face center
0	-1	1	-0.001414562726835	-0.018579946137910	-0.004428926517519	edge center
0	0	-1	0.000315189742674	0.012128861008564	0.021462177285547	face center
0	0	0	0.000000000000000	0.000000000000000	0.000000000000000	body center
0	0	1	-0.000315189742674	-0.012128861008564	-0.021462177285547	face center
0	1	-1	0.001414562726835	0.018579946137910	0.004428926517519	edge center
0	1	0	0.001099372984162	0.006451085129346	-0.017033250768028	face center
0	1	1	0.000784183241488	-0.005677775879218	-0.038495428053575	edge center
1	-1	-1	0.009406536016985	0.003969568631012	0.037328945095210	corner
1	-1	0	0.009091346274312	-0.008159292377552	0.015866767809663	edge center
1	-1	1	0.008776156531638	-0.020288153386115	-0.005595409475884	corner
1	0	-1	0.010505909001147	0.010420653760359	0.020295694327182	edge center
1	0	0	0.010190719258473	-0.001708207248205	-0.001166482958364	face center
1	0	1	0.009875529515800	-0.013837068256769	-0.022628660243911	edge center
1	1	-1	0.011605281985309	0.016871738889705	0.003262443559155	corner
1	1	0	0.011290092242635	0.004742877881141	-0.018199733726392	edge center
1	1	1	0.010974902499961	-0.007385983127423	-0.039661911011939	corner

B.2 Face-plane equations

Plane family	Unit normal	d (native)	Equation
ab faces (constant c coefficient)	(0.194092200808, 0.913177168821, 0.358379234791)	-1.882857350019305e-02	$n \cdot x + d = 0$
ab faces (constant c coefficient)	(0.194092200808, 0.913177168821, 0.358379234791)	1.882857350019305e-02	$n \cdot x + d = 0$
ac faces (constant b coefficient)	(0.089052594102, 0.866576769291, -0.491033948326)	1.405216690339142e-02	$n \cdot x + d = 0$
ac faces (constant b coefficient)	(0.089052594102, 0.866576769291, -0.491033948326)	-1.405216690339142e-02	$n \cdot x + d = 0$
bc faces (constant a coefficient)	(-0.995965190309, 0.083602162507, -0.032619290564)	-1.025436161905391e-02	$n \cdot x + d = 0$
bc faces (constant a coefficient)	(-0.995965190309, 0.083602162507, -0.032619290564)	1.025436161905391e-02	$n \cdot x + d = 0$

Appendix C. Scaling ledgers

C.1 Angular-rate scaling at R=0.10 m

rpm	omega (rad/s)	v (m/s)	aligned F (N)	E_rot (J)	ideal T (N)
1	1.047197551197e-01	1.047197551197e-02	7.859600438908e-07	5.483113556161e-05	1.745329251994e-04
9.5492965855	9.99999999986e-01	9.99999999986e-02	7.505365563465e-06	4.99999999986e-03	1.591549430914e-02
10	1.047197551197e+00	1.047197551197e-01	7.859600438908e-06	5.483113556161e-03	1.745329251994e-02
100	1.047197551197e+01	1.047197551197e+00	7.859600438908e-05	5.483113556161e-01	1.745329251994e+00
1000	1.047197551197e+02	1.047197551197e+01	7.859600438908e-04	5.483113556161e+01	1.745329251994e+02
10000	1.047197551197e+03	1.047197551197e+02	7.859600438908e-03	5.483113556161e+03	1.745329251994e+04
100000	1.047197551197e+04	1.047197551197e+03	7.859600438908e-02	5.483113556161e+05	1.745329251994e+06

C.2 Radius sweep at fixed omega=1 rad/s

R (m)	F (N)	v (m/s)	I shell	E_rot (J)	ideal T (N)
1.000000000000e-04	7.505365563476e-06	1.000000000000e-04	1.000000000000e-08	5.000000000000e-09	1.591549430919e-05
1.778279410039e-04	7.505365563476e-06	1.778279410039e-04	3.162277660168e-08	1.581138830084e-08	2.830219583062e-05
3.162277660168e-04	7.505365563476e-06	3.162277660168e-04	1.000000000000e-07	5.000000000000e-08	5.032921210449e-05
5.623413251903e-04	7.505365563476e-06	5.623413251903e-04	3.162277660168e-07	1.581138830084e-07	8.949940160889e-05
1.000000000000e-03	7.505365563476e-06	1.000000000000e-03	1.000000000000e-06	5.000000000000e-07	1.591549430919e-04
1.778279410039e-03	7.505365563476e-06	1.778279410039e-03	3.162277660168e-06	1.581138830084e-06	2.830219583062e-04
3.162277660168e-03	7.505365563476e-06	3.162277660168e-03	1.000000000000e-05	5.000000000000e-06	5.032921210449e-04
5.623413251903e-03	7.505365563476e-06	5.623413251903e-03	3.162277660168e-05	1.581138830084e-05	8.949940160889e-04
1.000000000000e-02	7.505365563476e-06	1.000000000000e-02	1.000000000000e-04	5.000000000000e-05	1.591549430919e-03
1.778279410039e-02	7.505365563476e-06	1.778279410039e-02	3.162277660168e-04	1.581138830084e-04	2.830219583062e-03
3.162277660168e-02	7.505365563476e-06	3.162277660168e-02	1.000000000000e-03	5.000000000000e-04	5.032921210449e-03
5.623413251903e-02	7.505365563476e-06	5.623413251903e-02	3.162277660168e-03	1.581138830084e-03	8.949940160889e-03
1.000000000000e-01	7.505365563476e-06	1.000000000000e-01	1.000000000000e-02	5.000000000000e-03	1.591549430919e-02

C.3 Radius sweep at fixed surface speed v=0.10 m/s

R (m)	omega (rad/s)	rpm	F (N)	E_rot shell (J)
1.000000000000e-04	1.000000000000e+03	9.549296585514e+03	7.505365563476e-03	5.000000000000e-03
1.778279410039e-04	5.623413251903e+02	5.369964096533e+03	4.220577217003e-03	5.000000000000e-03
3.162277660168e-04	3.162277660168e+02	3.019752726269e+03	2.373404985278e-03	5.000000000000e-03
5.623413251903e-04	1.778279410039e+02	1.698131749837e+03	1.334663704634e-03	5.000000000000e-03
1.000000000000e-03	1.000000000000e+02	9.549296585514e+02	7.505365563476e-04	5.000000000000e-03
1.778279410039e-03	5.623413251903e+01	5.369964096533e+02	4.220577217003e-04	5.000000000000e-03
3.162277660168e-03	3.162277660168e+01	3.019752726269e+02	2.373404985278e-04	5.000000000000e-03
5.623413251903e-03	1.778279410039e+01	1.698131749837e+02	1.334663704634e-04	5.000000000000e-03
1.000000000000e-02	1.000000000000e+01	9.549296585514e+01	7.505365563476e-05	5.000000000000e-03
1.778279410039e-02	5.623413251903e+00	5.369964096533e+01	4.220577217003e-05	5.000000000000e-03
3.162277660168e-02	3.162277660168e+00	3.019752726269e+01	2.373404985278e-05	5.000000000000e-03
5.623413251903e-02	1.778279410039e+00	1.698131749837e+01	1.334663704634e-05	5.000000000000e-03
1.000000000000e-01	1.000000000000e+00	9.549296585514e+00	7.505365563476e-06	5.000000000000e-03

Appendix D. Minimal machine reconstruction kernel

The following equations reconstruct the complete aligned tubular-rotor result from the machine-readable inputs.

```

Input vectors: c_hat, e0, A0, V_E; input matrix: G_phase; scalar: kappa_F

g = G_phase V_E

u(theta)=cos(theta)e0+sin(theta)A0
t(theta)=-sin(theta)e0+cos(theta)A0

dm=[M/(2 pi L)] dtheta dz

dF=-kappa_F dm [omega t(theta)^T g + R omega^2 t(theta)^T G_phase t(theta)]u(theta)

Pair theta with theta+pi: self-term cancels; translation term doubles

F=(kappa_F M omega/2)c_hat cross g

tau=0 for centered axial symmetry

J_revolution=F (2 pi/omega)

At fixed omega: F independent of R and L

At fixed surface speed v: F/M=K v/R

For thin wall M=2 pi R L rho t and F/L=2 pi K rho t v

```

Appendix E. Glossary

Term	Definition
Active mass	Rotor mass participating in the complete axisymmetric angular integration.
Angular phase rate	The angular speed omega; the calibrated force is linear in this rate.
CMB traversal axis	The locked ICRS unit vector c_hat defining the Earth-CMB reference direction.
Complete phase occupancy	Uniform mass at every circumferential angle so that all point-pair phases are present simultaneously.
G_phase	Symmetric physical response matrix that maps Earth velocity into the phase-weighted vector g.
Lateral force	The net force perpendicular to c_hat for the aligned reference rotor.
Radius of gyration	$k_r = \sqrt{I_p/M}$, the mass-weighted rotational radius controlling stored spin energy.
Surface speed	$v = R \omega$ at the outer rotating radius.
Tubular rotor	An axially extended set of complete circular cross-sections; it may be thin-wall, thick-wall, or modular.
Whirl critical speed	A bending-mode rotational speed at which imbalance and shaft deflection can resonate.

P. Dwayne Esterline, Principal Investigator

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